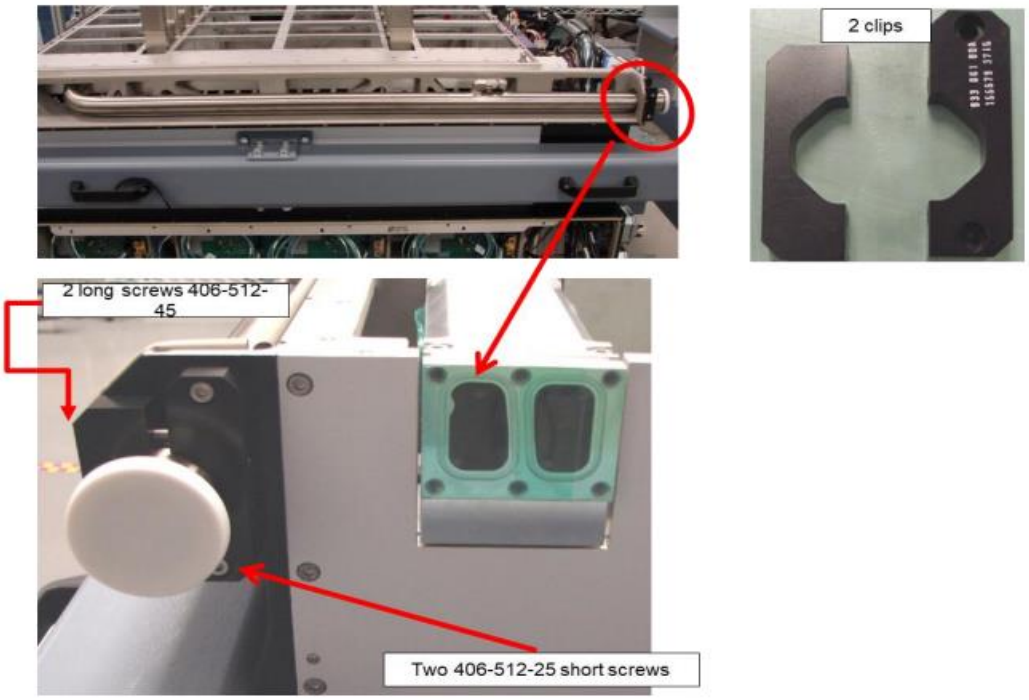
Build Overview:

1 **Return Manifold Install**
2 **Inlet Manifold Install**



CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
Manufacturing Engineer

Checked By:
IVAN NOWELL PANIQUE
Team Lead

Approved By:
SHERWIN ARIOSTE
Operations Manager

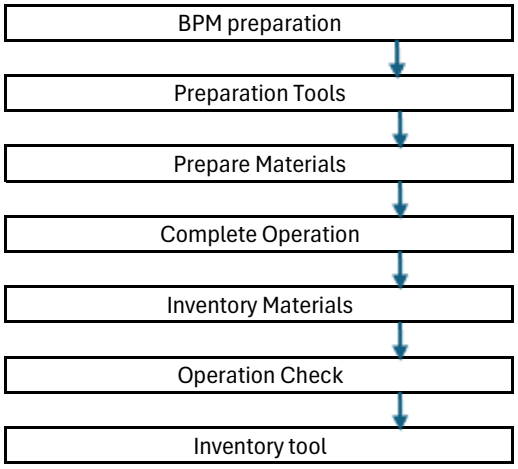
MAN: CERTIFIED TECHNICIANS.

MATERIAL:

NO.	PART NO.	DESCRIPTION	QTY.
1	638-030-00	DOW CORNING MOLYKOTE 111 COMPOUND GRAY 6G PACKAGE	1
2	361-607-06	O-RING 1-1/2IDX 1/16W E 0540-80	1
3	629-104-00	CAST STEEL OR IRON MANIFOLD 24 SLOT PIPE OUTLEST MISC	1
4	406-512-12	SCRW CP SKT GD M5X.8P/12 MM SS W/NYLOK	10
5	639-733-00	PLASTIC, CONDUCTIVE ELASTOMER GASKET, SEAL OR GASKET	1
6	479-283-02	FIT MALE CONNECTOR 3/8" TUBE OD X9/16-18 MALE SAE/MS STRGHT THRD. SS	1
7	633-061-00	ALUMINUM CLAMP MANIFOLD OUTLET BRACKET OR MISC ARTICLE	1
8	406-512-45	SCRW CP SKT HD M5X.8P/25 MM SS W/NYLOK	2
9	406-512-45	SCRW CAP SKT HD M5X.8P/45 MM L SS W/NYLOK	2
10	361-607-05	O-RING 1-3/8IDX 1/16W E 0540-80	2
11	629-106-00	CAST STEEL OR IRON MANIFOLD 24 SLOT END INLET MISC	1

METHOD:

Flow chart of operation steps



MACHINE:

TOOL NAME	TORQUE VALUE
4MM BALL HEAD TORQUE WRENCH	3.3 N.m
1/16 LONG SLEEVE TORQUE WRENCH	27 N.m
4MM HEX HEAD TORQUE WRENCH	3.3 N.m
4MM HEX BIT TORQUE WRENCH	3.3 N.m

MAN: CERTIFIED OPERATOR, TECHNICIAN

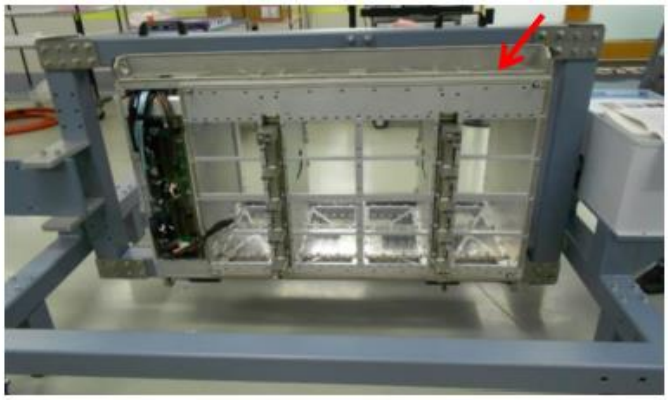
MATERIAL: N/A

MACHINE: N/A

METHOD: Operation steps

1.1 Rotate the T/H direction as shown to find the Manifold exit.

1.2 Carefully remove the plastic film on the Manifold outlet as shown in the figure,and check the circular groove for damage and debris.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

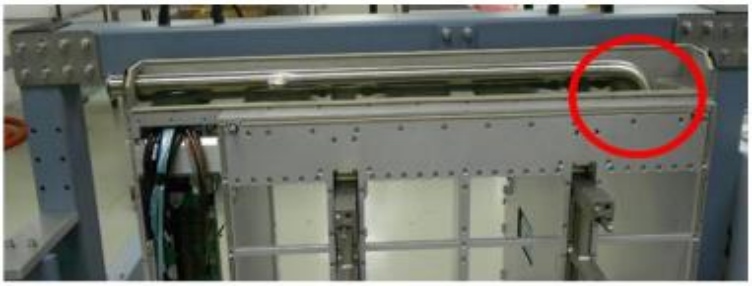
ITEM	PART NUMBER	DESCRIPTION	QTY
1	638-030-00	DOW CORNING MOLYKOTE 111 COMPOUND GRAY 6G PACKGE	1
2	361-607-06	O-RING 1-1/2IDX 1/16W E0540-80	1
3	629-104-00	CAST STEEL OR IRON MANIFOLD 24 SLOT PIPE OUTLET MISC	1
4	406-512-12	SCRW CP SKT HD M5X.8P/12 MM SS W/NYLOK	4

MACHINE: 4mm ball head torque wrench, 3.3 N·m.

METHOD:

1.3 Use Molykote grease (638-030-00) to lubricate the O-ring (361-607-06).
Put the lubricated O-ring into the outlet of Manifold.

1.4 Take a Return Pipe (629-104-00) and place it on the T/H frame as shown. Use 4 M5 screws (406-512-12) to fix the tube,
The tool is a 4mm ball head torque wrench with a torque value of 3.3 N·m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

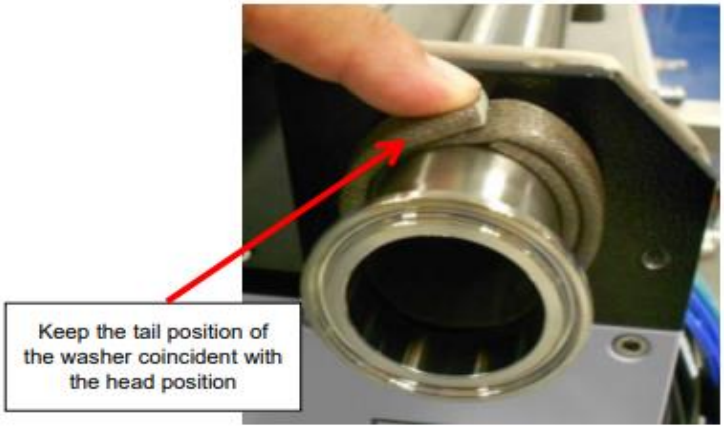
MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
5	639-733-00	PLASTIC, CONDUCTIVE ELASTOMER GASKET, SEAL OR GASKET	1

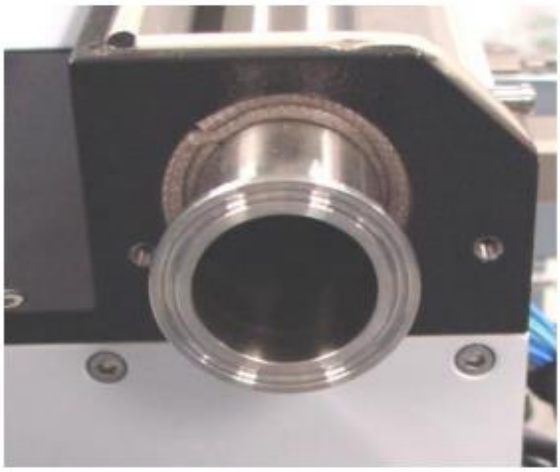
MACHINE: 3/16 socket wrench 0.19 N.m

METHOD:

1.5 Install a washer (639-733-00) around the pipe as shown in the figure to ensure that the tail position and the head position remain the same after the washer is wound twice.



1.6 Place the washer into the T/H frame as shown in the figure to ensure that the washer is fully pressed into the ring when it is inserted.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
6	479-283-02	FIT MALE CONNECTOR 3/8" TUBE OD X 9/16-18 MALE SAE/MS	1

MACHINE: 1/16 long sleeve torque wrench, 27 N·m.

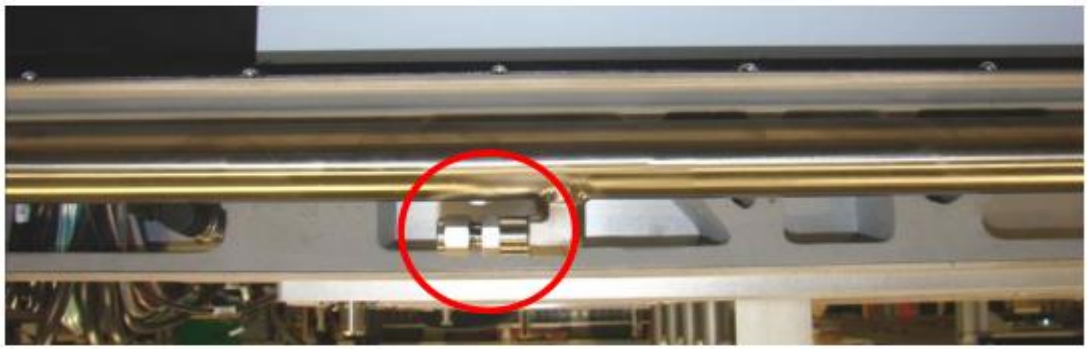
METHOD:

1.7 Install a joint sleeve (479-283-02).

a. Take a joint sleeve.

b. Use Molykote grease (638-030-00) to lubricate the O-ring on the joint sleeve before installation.

c. Install the joint sleeve on the pipe and tighten it by hand. Make a mark at the joint sleeve and elbow. Use an 11/16 long sleeve torque wrench to fix, The torque value is 27 N·m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
7	633-061-00	ALUMINUM CLAM MANIFOLD OUTLET BRACKET OR MISC ARTICLE	1
8	406-512-15	SCRW CP SKT HD M5X.8P/25 MM SS W/NYLOK	2
9	406-512-45	SCRW CAP SKT HD M5X.8P/45 MM L SS W/NYLOK	2

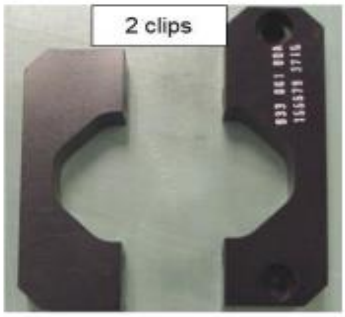
MACHINE: 4mm ball head torque wrench, 3.3 N·m.

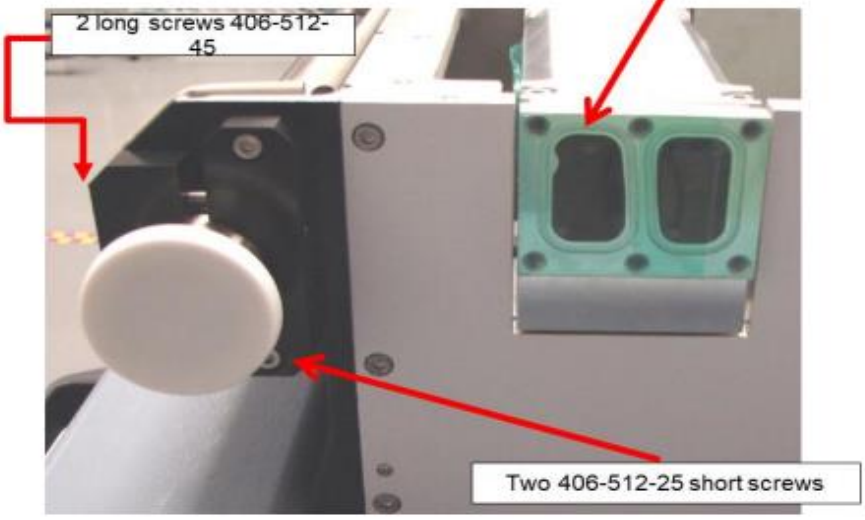
METHOD:

1.8 Install a Manifold tube clamp (633-061-00). At this time, it consists of 2 clips.

a. Install 2 clips as shown in the picture, use 2 M5 screws (406-512-25) and 2 M5 X 45 screws (406-512-45) to fix, The tool is a 4mm hex bit torque wrench with a torque value of 3.3 N·m.

b. Place the plastic cap on the tube.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

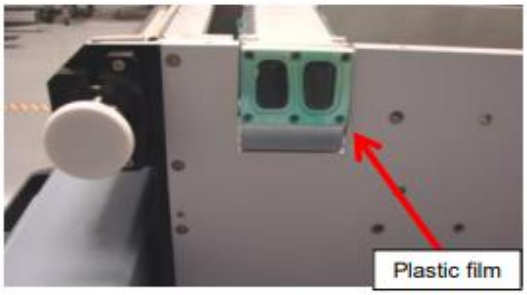
ITEM	PART NUMBER	DESCRIPTION	QTY
10	361-607-05	O-RING 1-3/8IDX 1/16W E0540-80	2
11	629-106-00	CAST STEEL OR IRON MANIFOLD 24 SLOT END INLET MISC	1
12	406-512-12	SCRW CP SKT HD M5X.8P/12 MM SS W/NYLOK	6

MACHINE: 4mm hexagonal head torque wrench,3.3 N·m.

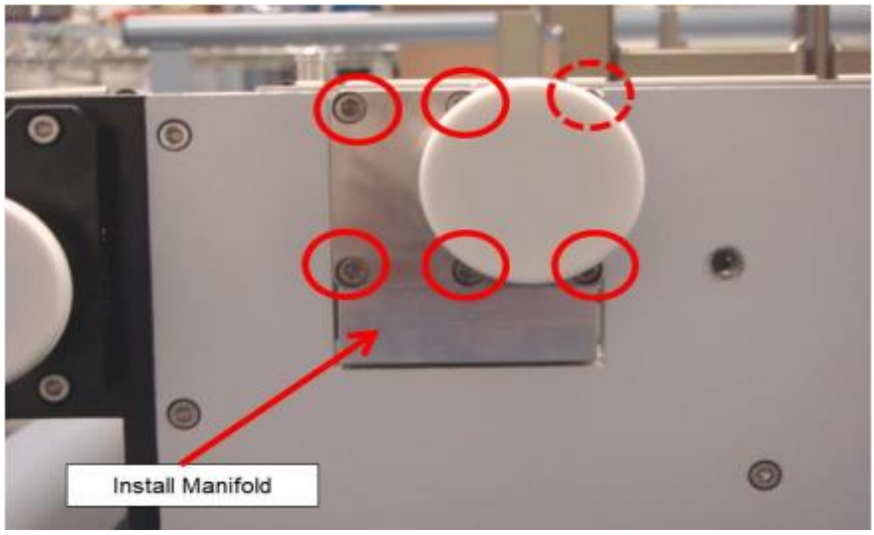
METHOD:

2.1 Carefully remove the T/H plastic protective film as shown, and check that the two O-ring grooves are not damaged and debris is on the surface

2.2 Before installing 2 O-rings (361-607-05), use Molykote grease (638-030-00) to lubricate the O-rings.



2.3 Install 1 Manifold (629-106-00), fix it with 6 M5 screws (406-512-12),

The tool is a 4mm hexagonal head torque wrench, The torque value is 3.3 N·m. Put on the plastic cover.

2.4 Note that when installing Manifold, the nozzle direction should be on the right.